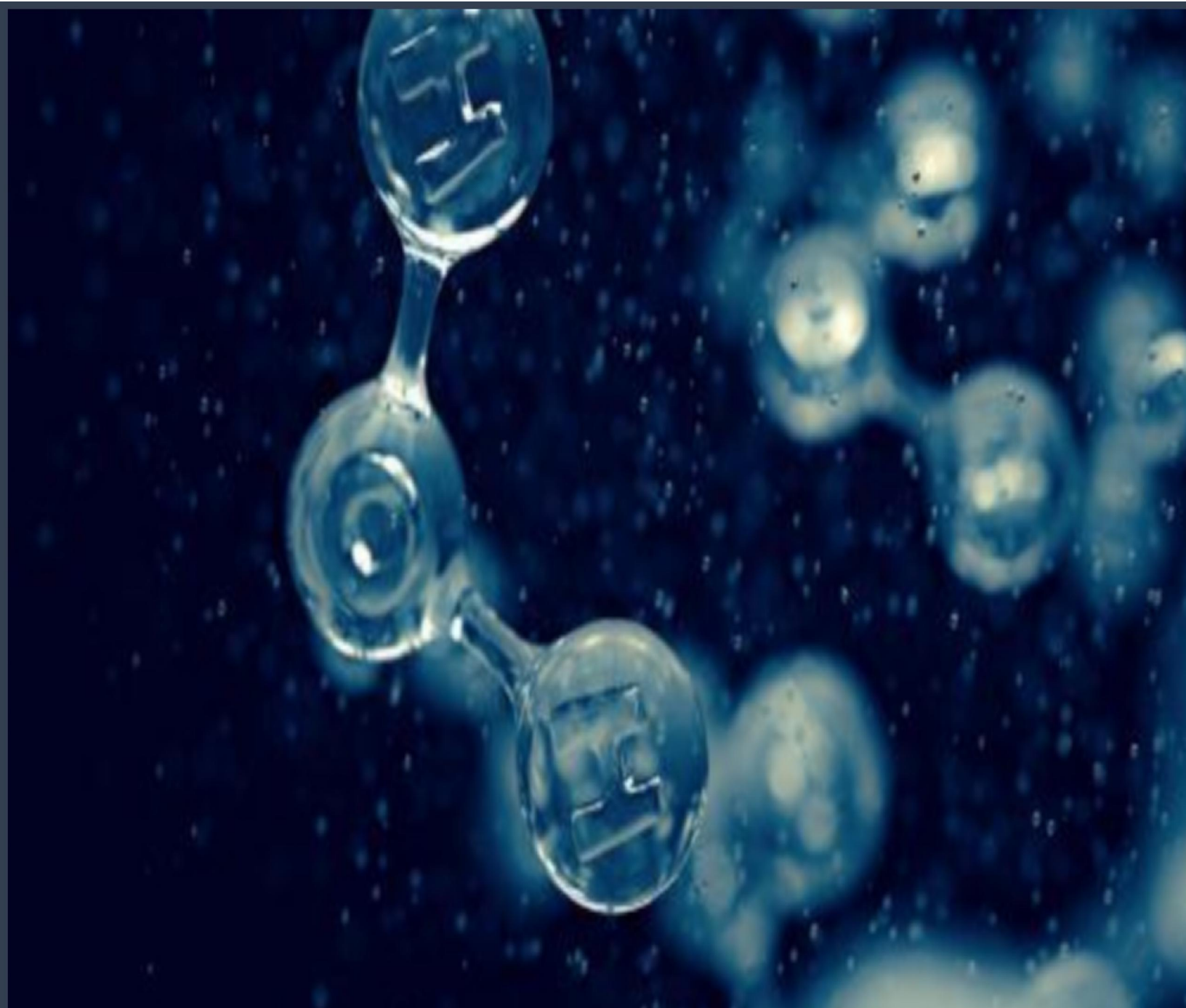




Water

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Objective of the session

After completing this session students are expected to know :

- The Molecular Makeup of Water
- Water Sources and Intake
- Water Requirement
- Biological Role of Water
- Distribution of Body Water
- Deficiency and Toxicity of water



Introduction

Space scientists are looking for water on Mars in their search for life because we all know right from our elementary school days that water is important for all living systems.

Water makes up 60-75% of human body weight.

An individual can survive a month without food but can survive only 3 days without water, as a loss of just 4% of total body water leads to severe dehydration, and a loss of 15% is fatal.

We need to understand the question: What makes water so essential for survival?

Viewed from space, the most striking feature of Earth is the water. Seventy-five per cent of the Earth's surface is covered with its liquid and frozen forms, It fills the sky with clouds. Water is practically everywhere on Earth from inside the planet's rocky crust to inside the cells of the human body.

This photo-like view of Earth is based largely on observations from MODIS, the Moderate Resolution Imaging Spectroradiometer, on NASA's Terra satellite. Image Credit: NASA

Pure water is the world's first and foremost medicine (Slovakian proverb)

The Molecular Makeup of Water

- Water is not only an inert solvent but is the substrate for many cellular reactions.
- In order to understand the biochemistry of nutrients, one first needs to understand the chemistry of water.
- Water is a molecule composed of two small positively charged hydrogen atoms and one large negatively charged oxygen atom with the molecular formula H_2O .
- When the hydrogen atoms bind to oxygen, they are held together by strong covalent bonds.
- The structure of water shows two bonding pairs of electrons and two non-bonding pairs of electrons.
- The four pairs repel one another, forming a tetrahedral pattern making the molecule "angular" or "nonlinear", with an H-O-H bond angle of free water being 104.5° .
- The two electrons in each oxygen-hydrogen bond are not shared equally.

- They are more strongly attracted to the oxygen atom.

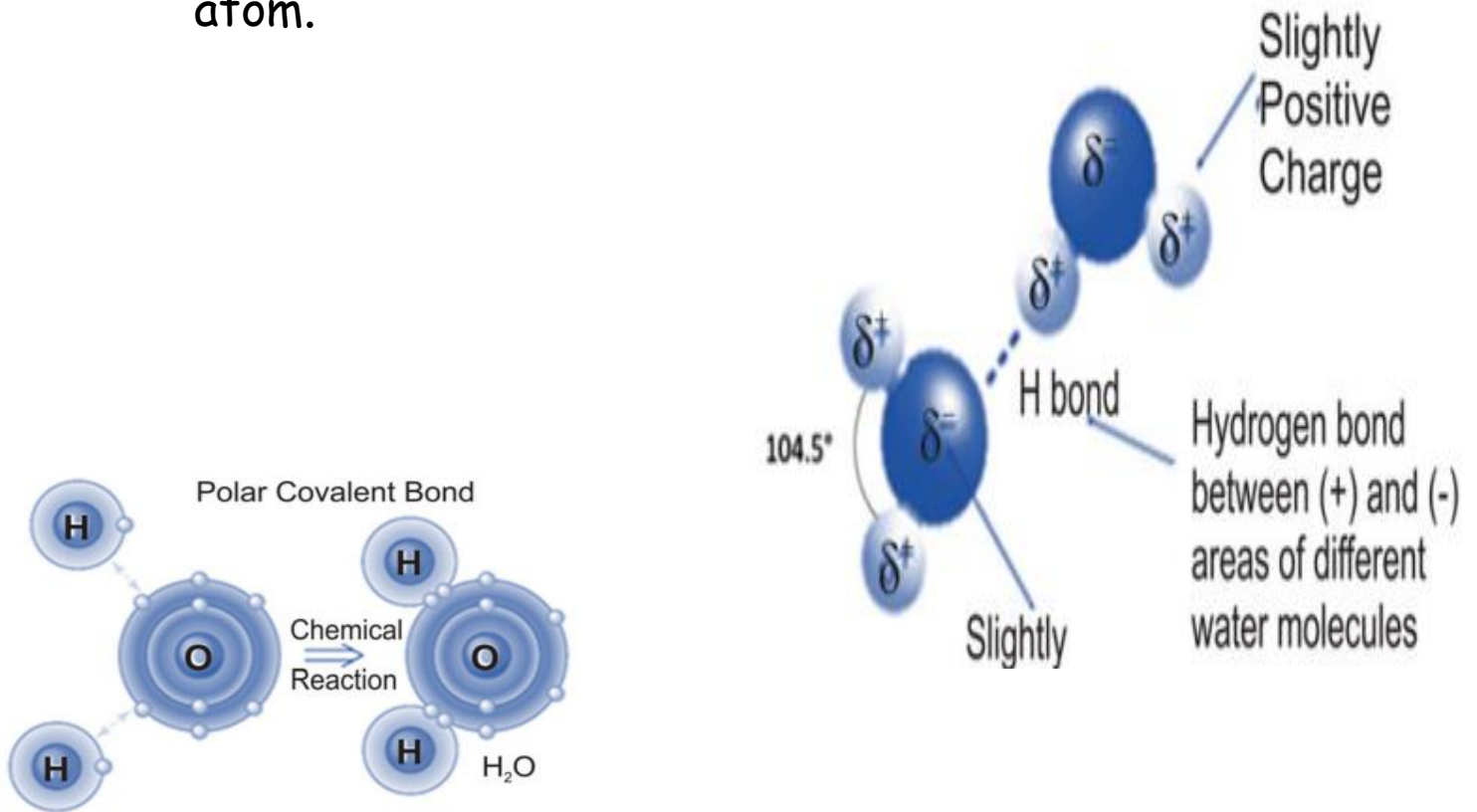


Fig.1:Chemistry of water. Water molecules are made of two hydrogens and one oxygen. These atoms react forming a polar covalent bond that creates a charge difference and asymmetry in the molecular structure. This leads to the formation of a hydrogen bond between water and other polar molecules, including between two water molecules

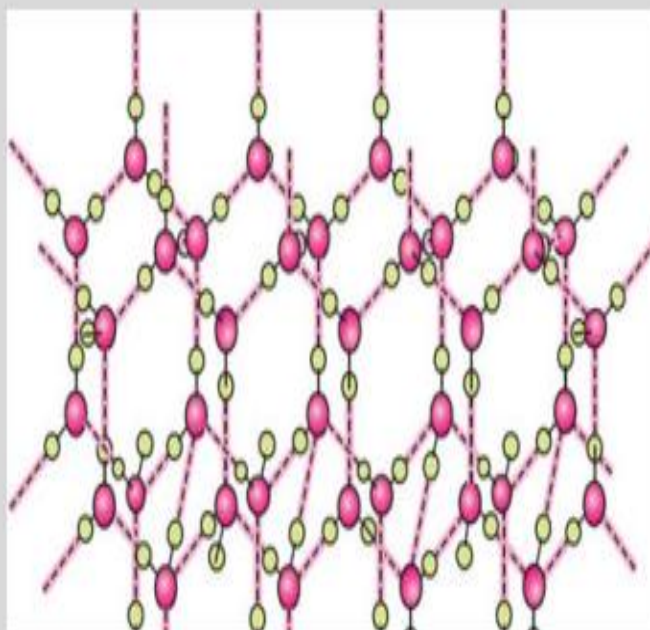
- This creates an asymmetrical molecule with a positive charge (the hydrogen atom) on one side and a negative charge (the oxygen atom) on the other side.
- This charge differential is called polarity.
- A hydrogen bond forms between a non-bonding pair of electrons on the oxygen atom of one water molecule and the hydrogen atom ("positive end") of another water molecule.
- The hydrogen bond is about 10 times weaker than a single covalent bond.

Floating ice saves marine life



Floating glaciers in sea. (Source: <https://www.flickr.com/photos/arabani/5136296355/>)

A water molecule has four hydrogen bonds, each of which points to the oxygen atom of an adjacent water molecule, and these four nearby hydrogen-bonded oxygen atoms occupy the vertices of a tetrahedron. In water, the molecules can move more freely and pack closely together. However, unlike other solvents which tend to become denser on freezing, when the temperature dips below 4°C , water expands and ice is formed. In ice crystals, formation of the more open and rigid hydrogen bonds occurs, in which each water molecule is hydrogen-bonded tightly to four others, causing this expansion. As a result, ice has a lower density (0.9168 g ml^{-1}) than water ($0.9998 \cong 1.0\text{ g ml}^{-1}$), and hence it floats and water freezes from the top down. This is termed as the “anomalous nature of water”.



Structure of water in which water molecules form hydrogen bonds, forming an open hexagonal lattice

A covering of ice on a water body creates an insulation such that the animals underneath can survive the extreme cold. The water underneath the ice does not freeze, allowing for marine life to survive even in the freezing arctic and polar regions.

Water Sources and Intake

- Total water intake for a person includes the water consumed as food and beverages as well as the small volumes of water created by oxidation of food (metabolic water) and breakdown of body tissue.
- The amount of metabolic water for an average adult ranges from about 350 to 400 mL/day, but even this can vary considerably.
- Humans ingest water as plain drinking water, as beverages of varied content, and in food. Water in food can be inherent or added during preparation.

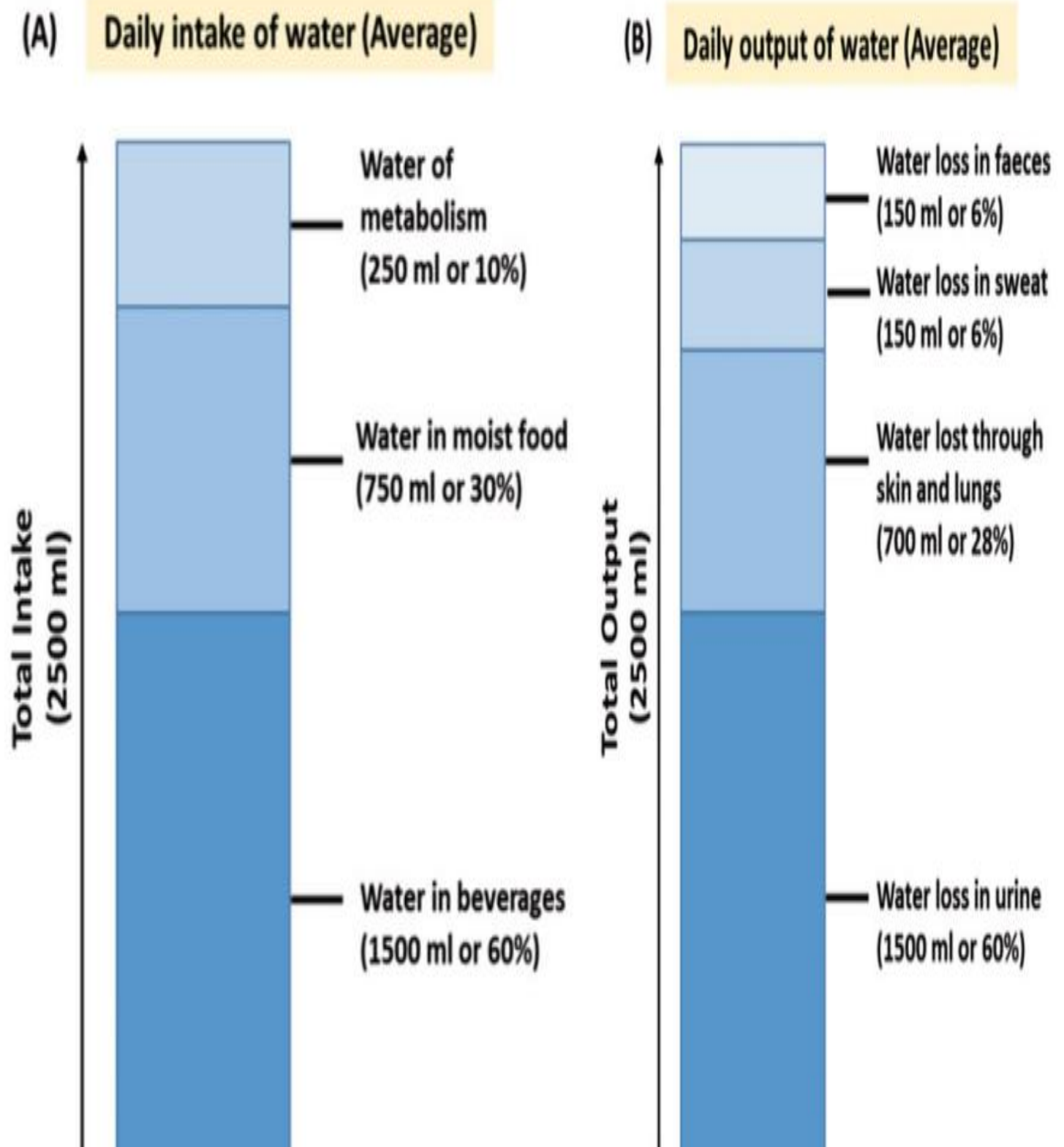


Fig.2: (A) The daily average intake of water in the form of beverages and food along with the small amounts of water produced during metabolism should compensate for the daily average losses of water. (B) In humans, the losses are primarily through urine, but significant amounts are also lost through skin and respired out through lungs. Small amounts are also lost through faeces and sweat.

Water content of beverages varies.

- Plain drinking water and diet soft drinks are 100% water, whereas coffee and tea are 99.5% water, and a sports drink has 95% water.
- Water content in fruit juices varies from 90 to 94% water.
- Skimmed milk, toned milk with 2% fat, and whole milk contain 91%, 89%, and 87% of water, respectively.
- Some beverages have a diuretic effect on the human body and hence though consumption leads to fluid intake; this is counterbalanced by excessive urinary losses that ultimately may lead to dehydration.
- **diuretic effect:** refers to the increased production and excretion of urine by the kidneys.
- Substances that cause this effect are known as **diuretics**.
- These substances promote the removal of water and electrolytes (such as sodium and potassium) from the body, primarily through urine.
- The diuretic effect can be caused by certain foods, beverages, medications, and medical conditions.

● How the Diuretic Effect Works

- ✧ **Kidney Function:** The kidneys regulate fluid balance in the body by filtering blood, reabsorbing necessary substances, and excreting waste products and excess water as urine.
- ✧ Diuretics work by altering the way the kidneys handle sodium. Most diuretics inhibit sodium reabsorption, leading to more sodium being excreted in the urine. Since water follows sodium, increased sodium excretion results in increased water excretion.

● Types of Diuretics:

- I. **Loop Diuretics:** These act on the **loop of Henle** in the kidneys to inhibit sodium, chloride, and potassium reabsorption, leading to significant urine production. They are commonly used to treat conditions like **heart failure and edema**.
- II. **Thiazide Diuretics:** These act on the **distal convoluted tubule** in the kidneys and are often used to treat **high blood pressure** by reducing blood volume.

III. Potassium-Sparing Diuretics: These reduce sodium reabsorption while conserving potassium. They are often used in combination with other diuretics to prevent potassium loss.

IV. Osmotic Diuretics: These increase the osmolarity of the blood, drawing water into the blood from tissues, and then excreting it through the kidneys. They are used in specific medical situations, like reducing **intracranial pressure**.

● Common Substances with Diuretic Effects

✧ **Caffeine:** Found in coffee, tea, soda, and energy drinks, caffeine has mild diuretic effects. It increases urine production by inhibiting the reabsorption of sodium and water in the kidneys. However, the effect is usually mild, especially in people who regularly consume caffeine.

✧ **Alcohol:** Alcohol inhibits the release of vasopressin (antidiuretic hormone or **ADH**), a hormone that promotes water reabsorption in the kidneys. This leads to increased urine production and can result in dehydration if not offset by adequate water intake.

- ✧ **Certain-Foods**: Some foods, like parsley, dandelion greens, and watermelon, have natural diuretic properties, promoting increased urination.

● Clinical Uses of Diuretics

- ✧ **Hypertension(HighBloodPressure)**: Diuretics are commonly prescribed to lower blood pressure by reducing blood volume, which in turn decreases the pressure on blood vessel walls.
- ✧ **Edema**: Conditions like heart failure, liver cirrhosis, and kidney disease can lead to fluid retention (edema). Diuretics help reduce this fluid buildup by promoting its excretion.
- ✧ **HeartFailure**: In patients with heart failure, diuretics are used to reduce the workload on the heart by decreasing the volume of fluid the heart must pump.

● Side Effects and Considerations

- ✧ **Dehydration**: Excessive use of diuretics can lead to dehydration, characterized by symptoms such as dry mouth, dizziness, and reduced urine output.
- ✧ **Electrolyte Imbalance**: The loss of essential electrolytes, particularly potassium, can cause muscle cramps, weakness, and, in severe cases, heart arrhythmias.

- ✧ **Increased Thirst:** As the body loses water, it may trigger a feeling of thirst, prompting the person to drink more fluids.
- ✧ **Frequent Urination:** One of the most noticeable effects of diuretics is the need to urinate more frequently, which can be inconvenient for some people.

Water Requirement

- The amount of water required by an individual should equal losses from the body and prevent adverse effects such as dehydration.
- For a sedentary to moderately active individual under temperate conditions, water is lost from the body via urine, faeces, respiration, and evaporation.
- However, with increased physical activity and in tropical and hot environmental conditions, loss through sweat is also significant.
- Water is also lost through the skin via trans-epidermal diffusion followed by evaporation, and also from the respiratory tract as water vapour in the expired air.

- Both these losses are collectively referred to as insensible water loss which directly correlates with metabolic heat dissipation.
- Environmental temperature and humidity, altitude, volume of air inspired, air currents, clothing, blood circulation through skin, and water content of the body can all affect insensible water loss.
- The minimal amount of fluid loss that can occur is referred to as the **obligatory water loss** and the amount of water needed to replace such losses is the **absolute requirement**.
- **Obligatory water loss** is the minimum amount of water that must be excreted daily to remove waste products and maintain essential physiological functions.
- This includes water lost through urine, feces, and insensible processes like breathing and evaporation from the skin.
- However, even this is variable as a variety of factors can affect obligatory loss.

● Factors Affecting Obligatory Water Loss

- ✧ **Diet:** A diet high in protein or salt increases the amount of solutes the kidneys must excrete, potentially increasing obligatory water loss.
 - ✧ **Health Conditions:** Conditions like fever, diarrhea, or diabetes can increase obligatory water loss due to higher metabolic activity or loss of fluids through other means.
 - ✧ **Environment:** Dry or cold environments may increase insensible water loss, while hot environments increase sweating (though sweating is not considered part of obligatory loss).
 - ✧ **Age:** Infants and elderly individuals may have different obligatory water loss rates due to variations in metabolism, kidney function, and body surface area relative to body mass.
- Given the multitude of intra- and inter-individual factors that influence water requirements, it becomes not only impossible but also impractical to establish a general water requirement that would ensure adequate hydration and optimal health to all.
 - Thus, the WHO has stated that an estimated average requirement (EAR), and therefore a

recommended dietary allowance (RDA), cannot be established.

- Hence, an adequate intake (AI) is given as the reference value for water intake.
- Another complication that arises when trying to compute water intake is the fact that water as pure H₂O is never consumed.
- Even the purest of water contains salts, particularly electrolytes.
- Hence, the requirement of water is also coupled with electrolyte balance in the body.
- Age- and gender-specific adequate intakes (AI) for water were established in 2004 by the Food and Nutrition Board (USA).

Adequate intakes for total water		
Age group	AI (l/day)	
0–6 months	0.7	
7–12 months	0.7	
1–3 years	1.3	
4–8 years	1.7	
	Male	Female
9–13 years	2.4	2.1
14–18 years	3.3	2.3
19–70+ years	3.7	2.7

Fig .4:Adequate intakes (AI) of water for different age groups

● Factors Influencing Water Needs

- ✧ **Physical Activity**: Increased physical activity, especially in hot or humid environments, can significantly raise water needs due to increased sweat loss.
- ✧ **Climate**: Hot, humid, or cold climates can affect hydration needs. In hot climates, more water is needed to replace fluid lost through sweating, while in cold climates, fluid loss can occur through respiration.
- ✧ **Health Status**: Conditions such as fever, diarrhea, vomiting, and infections can increase water requirements. Pregnant or breastfeeding women also need more water.
- ✧ **Age**: Children and elderly individuals may have different hydration needs. Children require adequate water relative to their body size, while the elderly may need to be mindful of maintaining hydration due to a reduced sense of thirst.

● Practical Tips

- ✧ **Listen to Your Body:** Thirst is a natural signal that your body needs water. Drinking when you're thirsty and maintaining a balance of fluid intake throughout the day helps ensure proper hydration.
- ✧ **Monitor Urine Color:** Light yellow or clear urine usually indicates adequate hydration, while dark yellow urine may suggest that you need to drink more water.
- ✧ **Adjust intake Based on Activity and Environment:**

Increase water intake during periods of physical exertion, hot weather, or illness to compensate for additional fluid loss.

Biological Role of Water

- Water plays the role of the universal solvent, while also acting as a metabolite, pH buffer, and temperature regulator. Water is also essential for maintaining molecular and cellular structures.
- It further acts as a lubricant in the joints, eyeballs, etc. and also has protective roles in the form of secretions.

Distribution of Body Water

- Water makes up about 60% of an adult's total body mass, and it plays a key role in many body functions.
- We have also discussed that water is a universal solvent and serves as a transport medium.
- The body does not have deionised pure H₂O; all water in the body contains ions and other organic solutes.
- Amongst the ions it is electrolytes like sodium, chloride, and potassium that help to equilibrate and distribute water throughout the body.
- They thus help maintain osmolarity and tonicity of cells and tissues within the body. Hence in biological terms the word fluid is preferably used instead of water.
- The total body water is primarily divided into the fluid present within the cells and the fluid present outside the cells, termed intracellular and extracellular fluid, respectively .

- Most of the water in the body is present inside the cells. The fluid outside the cells is again classified based on its localisation in the body.
- The fluid that bathes the cells within a tissue is called interstitial fluid, which is in equilibrium and taken up through open-ended lymphatic vessels into the lymph.
- The lymphatic circulation drains the lymph into the systemic circulation which contains plasma, the fluid component of blood.
- This plasma is pumped throughout the body via the cardiac-driven circulatory system.
- The transcellular fluids are all other secretory and excretory fluids and include cerebrospinal fluid, synovial fluid, ocular fluid, gastrointestinal tract secretions, and urine and sweat.
- The volume and the ionic composition of all body fluids are tightly regulated by both hormonal and neural control.

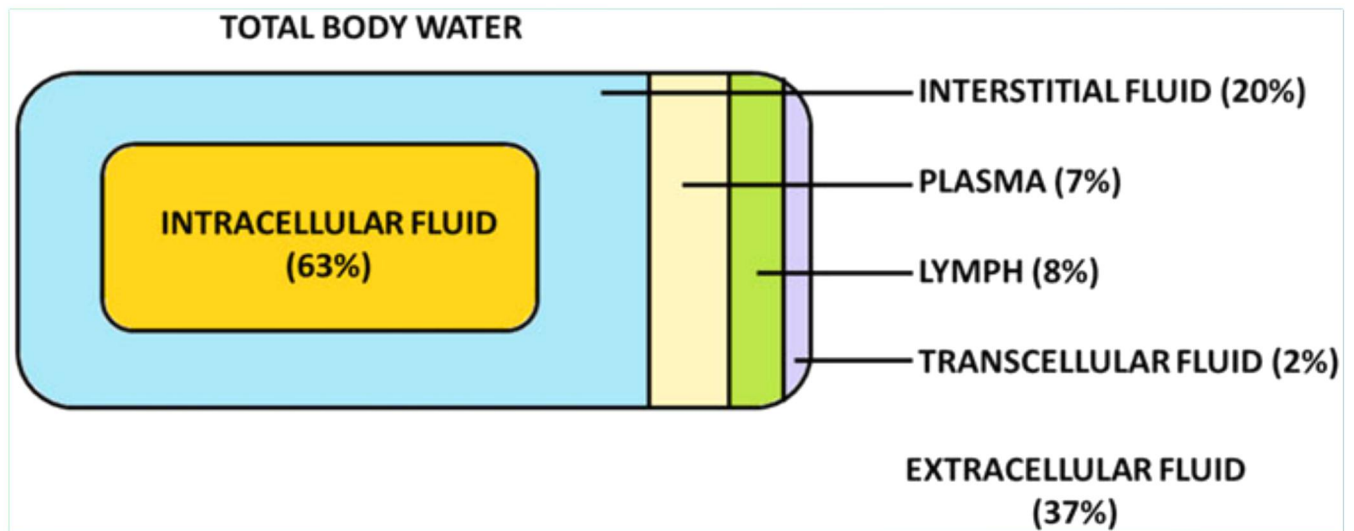


Fig 4: Distribution of total body water. Intracellular fluids are present inside the cells while interstitial fluid surrounds the cell within a tissue. Plasma and lymph are the major circulatory fluids. Transcellular fluids constitute a very small percentage of the total body water, including important fluids like CSF, synovial fluids, and excretory fluid like urine .

Deficiency and Toxicity

- Fluid-disease relationships have been studied extensively. Most studies have considered various combinations of variables including dehydration, hyper hydration, fluid volume consumed, and types of beverages, in relation to the absence, presence, or treatment of certain diseases or conditions.
- Dehydration or a decrease in total body water has been linked to an increased risk of contracting many diseases such as constipation, kidney stones, urinary tract infections, dental disease, bronchopulmonary disorders, and impaired cognition.
- A relationship between a high fluid intake and decreased risk of a variety of maladies including urinary tract stones, colon and urinary tract cancer, and mitral valve prolapse has been shown.
- However, as with most epidemiology studies, known and unknown interfering variables make it impossible to determine any definitive cause-effect relationship between fluid intake and disease susceptibility and/or prognosis.

- The amount of fluid necessary to maintain hydration should be the primary concern rather than determining fluid intake necessary to treat or decrease risk of certain diseases or disorders.
- However, fluid volume in the human body is intimately connected to electrolyte balance.
- Variation in fluid volumes is often reflected as electrolyte imbalance, and this causes most of the adverse symptoms associated with fluid volume deficiency or excess.

Dehydration

- Dehydration happens when the amount of fluid required by the body is far greater than the amount available.
- Based on the degree of imbalance in required water and available water, dehydration can be mild, moderate, or severe.
- Dehydration can be caused by either excessively low intakes of fluids and/or electrolytes or the acute loss of fluids.

- Decreased intake of water can occur due to non-availability like in drought conditions, or no accessibility to a water source (lost in a desert, hurt and lost, etc.).
- In these conditions, the only source of water supply is metabolic water.
- In such conditions normally the body's homeostatic mechanisms ensure adequate extracellular water content at the expense of the intracellular fluid, leading to cellular dehydration.
- One can also lose more water than usual if the person is suffering from fever, diarrhoea, vomiting, and excessive sweating.
- Excessive urinary losses can also occur in conditions like diabetes (mellitus and insipidus) and ingestion of drugs and beverages that are diuretics.
- In all the above cases, the dehydration is called mixed dehydration as it is usually characterized by both water and electrolyte depletion.
- The first outward reflex of dehydration is the feeling of thirst followed by the typical symptoms, first observed as a decrease in epidermal and ocular

water (areas where water loss through evaporation is high) leading to parched skin identified by the skin pinch test and sunken eyes .

Parameters	No dehydration	Some dehydration	Severe dehydration
Appearance	Person appears alert	Person appears irritated and restless	Person appears lethargic or unconscious
Eyes	Eyes are normal	Eyes are sunken	Eyes are very sunken
Thirst	Drinks normally not thirsty	Person is very thirsty and drinks water eagerly	Person is not able to drink water or drinks poorly
Skin pinch	Skin goes back quickly within 1 s	Skin goes back slowly in about 1s	Skin goes back very slowly in more than 2 s

Table 1 :WHO guidelines of clinical symptoms that can be used for infield identification of mild, moderate, or severe dehydration.

- Apart from the classification based solely on fluid volumes, dehydration is also classified as hyperosmolal, isoosmolal, and hypoosmolal.
- This is based on the changes in total body water, extracellular fluid (ECF) volumes, and electrolyte concentrations in intracellular fluid (ICF) versus ECF.
- Extreme water deprivation with no loss of electrolytes stimulates the secretion of antidiuretic hormone which triggers water reabsorption in the kidneys and at the same time both salts and water are dragged out of the cells to compensate water loss of ECF, leading to intracellular dehydration.
- When both salts and water are lost equally, the dehydration is isoosmolal.
- When salt loss is higher than water loss, the body's compensatory mechanisms lead to an increase in ICF as compared to ECF.

Dehydration type	Osmolarity	TBW	ICF	ECF	Causes	Xrc-
Isoosmolal	Normal(280-300 mOsm/L)	Decreased	Unchanged	Decreased	Diarrhea Vomiting Burns Hemorrhage	Hypotension Tachycardia Dry skin Normal Na ⁺ Urine Output: Oliguria
Hypoosmolal	Decreased (<280 mOsm/L)	Slightly decreased or normal	Increased	Decreased	Inappropriate Rehydration, Certain diuretics, adrenal insufficiency, Renal losses	Hyponatremia, cellular edema, neurological symptoms
Hyperosmolar	Increased (>295 mOsm/L)	Decreased	Decreased	Decreased but hyperosmolar	Fever, diabetes insipidus, severe hyperglycemia	Hypernatremia, thirst, neurological symptoms

Table.2: Types of dehydration

● Overhydration

- Water intoxication synonymously called water poisoning or dilutional hyponatraemia (decreased sodium levels in blood) is caused by overhydration in a short time period without giving the body proper electrolyte intake.
- It rarely results from overconsumption of just water but is usually a combination of excessive fluid intake and an increased secretion of vasopressin (antidiuretic hormone).
- Vasopressin secretion increases in periods of physical stress like during a marathon, as the body is adapted to conserve water. In such situations drinking excessive quantities of plain water with poor electrolyte balance will lead to overhydration.
- Based on the relationship with electrolyte levels, overhydration is class-fied as isotonic, hypertonic, or hypotonic overhydration.
- Hypovolemia also called isotonic overhydration occurs when ECF volume increases, and this can lead to circulatory overload and oedema.

- Excessive intake of NaCl can cause compensatory water retention, and this is called hypertonic overhydration.
- True water intoxication is the excessive consumption of pure water which leads to hyponatraemia and is called hypotonic overhydration .
- Water intoxication can cause death. Although it is rare, there are many tragic examples of death by water.
- In 2007, a 28-year-old California woman died after competing in a radio station's on-air water-drinking contest. After drinking some 6 L of water in 3 h in the "Hold Your Wee for a Wii" contest, Jennifer Strange vomited, went home with a splitting headache, and died from so-called water intoxication.
- A 2005 study in the New England Journal of Medicine found that close to one-sixth of marathon runners develop some degree of hyponatraemia, or dilution of the blood caused by drinking too much water.

- Symptoms of water intoxication appear after consumption of more than 3-4 L of water in a few hours.
- Most of the clinical symptoms are related to the associated severe hyponatraemia. These include head pain, cramping, spasms, or weakness in muscles, nausea or vomiting, drowsiness, and fatigue.
- Cases of acute water intoxication leading to water poisoning can also cause seizures and loss of consciousness and if not treated immediately can be fatal.

Overhydration type	Osmolarity	TBW	ICF	ECF	Causes	Xrc-
Isotonic Overhydration	Normal (280-295 mOsm/L)	Increased	Unchanged	Increased	Excessive isotonic IV fluids, kidney failure, heart failure, Liver Cirrhosis, Long-Term Corticosteroid Use	Elevated BP, pulmonary edema,
Hypotonic Overhydration	Decreased (<280 mOsm/L)	Increased	Increased	Increased but diluted	Excessive water intake, SIADH, kidney disorders, Hypotonic IV Fluids	Hyponatremia, cerebral edema, muscle cramps
Hypertonic Overhydration	Increased (>295 mOsm/L)	Increased	Decreased	Increased with hyponatremia	Hypertonic fluid administration, hyperaldosteronism (Excessive aldosterone production), sea water ingestion	Hypernatremia, thirst, neurological symptoms

Table 3: Types of overhydration



Symptoms of overhydration :According to the research, most cases of water poisoning do not result from simply drinking too much water. It is usually a combination of excessive fluid intake and increased secretion of vasopressin that is produced by the hypothalamus and secreted into the bloodstream by the posterior pituitary gland. Vasopressin regulates water conservation through the kidneys. Its secretion increases in periods of physical stress—during a marathon, for example—and may cause the body to conserve water even if a person is drinking excessive quantities. Every hour, a healthy kidney at rest can excrete 800–1000 ml of water and therefore a person can drink water at a rate of 800–1000 ml/h without experiencing a net gain in water. If that same person is running a marathon, however, the stress of the situation will increase vasopressin levels, reducing the kidney's excretion capacity to as low as 100 ml/h. Drinking 800–1000 ml of water per hour under these conditions can potentially lead to a net gain in water, even with considerable sweating. This in turn provokes disturbances in electrolyte balance, resulting in a rapid decrease in serum sodium concentration and eventual death.



Thank You